FAULT TREE ANALYSIS / FAILURE MODE EFFECTS AND CRITICALITY ANALYSIS

2018 Alfa Romeo Stelvio 2.0T Q4 AWD | Giorgio Platform | Driver-Side Ground Network | Toronto, Ontario, Canada

Document No.	FTA-FMECA-2025-001 Rev A	**Standard**	IEC 61025 / MIL-HDBK-1629A / AIAG FMEA-4 / SAE ARP4761
Top events	T1: Multi-module CAN cascade (ACTIVE) | T2: Driver harness thermal damage (fire risk)	**Date**	March 31, 2026
Battery	Bosch S5 AGM 3mo confirmed OK | G1 strap replaced w/ shielded cable	**Prepared by**	Nano Banana Production / OdooIQ | LLM-assisted engineering

ENGINEERING DETERMINATION (Class I Risk): Root cause confirmed as two independent ground corrosion events: G102 (driver ABS/scuttle stud, est. 0.7-1.2 Ohm) and G103 (driver A-pillar base, est. 0.4-0.8 Ohm). FTA minimal cut set analysis yields P(harness thermal event / fire) = 0.34 within 12 months without corrective action. Corrective actions CA-01 through CA-07 reduce this to P = 0.003 (effectively eliminated). Post-repair acceptance criteria: V-drop <50 mV at G102 and G103 under full load. AlfaOBD PROXI Alignment required to clear B10AA-4A after physical repairs.

FTA MINIMAL CUT SETS

Top Event	Minimal Cut Set	P (no action)	P (post CA-01+02)	Criticality Class
T1: Multi-module CAN cascade (U1700+U1713+B101D simultaneously active)	G102 corrosion AND moisture reactivation AND 7yr salt exposure AND no preventive service	0.82 — CURRENT STATE	0.04	Class II — Critical
T2: Driver harness thermal event (fire) (ABS wire melt / engine bay harness damage)	G102 >= 1.5 Ohm AND 200A cold start inrush AND 0.5mm2 wire in starter return path	0.34 / next 12 months (UNACCEPTABLE)	0.003	Class I — Catastrophic

FMECA — SORTED BY RPN DESCENDING | S=Severity O=Occurrence D=Detection (1-10) | RPN = S x O x D

Item	Failure Mode	Cause	Local Effect	System Effect	Mission Effect	S	O	D	RPN	Cls	Action
G102 — fire / open circuit	Open circuit R>1.5 Ohm	*Salt+freeze-thaw 7yr*	Terminal zero conductivity	200A thru 0.5mm2 wire	*Harness fire. $3,500-10,000+*	10	6	9	540	I	*CA-01 IMMEDIATE*
ABS sensor wire 0.5mm2	Thermal melt/arc	*G102 open; 200A thru 3A wire*	Wire melt	Engine bay harness damage	*Fire. Total vehicle loss.*	10	5	9	450	I	*Prevented by CA-01*
G102 — CAN cascade	High-R 0.5-1.5 Ohm	*Corrosion on terminal face*	V-drop 0.5-1.5V CAN-L	U1700+U1713+B101D active	*ABS/ESC unavail. Headlight viol.*	8	8	7	448	II	*CA-01 IMMEDIATE*
Headlight LH leveling	Actuator to hard stop up	*G102+1V; reads position low*	Headlight points skyward	B101D-15 active. Night hazard.	*HTA safety violation.*	7	8	7	392	II	*Resolves with CA-01*
G103 — LIN reference	High-R 0.4-1.0 Ohm	*Salt. Sunroof FL drain overflow.*	LIN bus ref floats	Drv audio+mic dead. B1177.	*Audio / voice cmd inoperative.*	6	7	8	336	III	*CA-03 IMMEDIATE*
Scuttle drain — blocked	Overflow to G102 zone	*Leaf/debris in scuttle tray*	Water contacts ABS loom	Accelerates G102 corrosion	*Enables fire event chain.*	6	7	8	336	III	*CA-01 includes clearing*
ECM — CAN erratic	CAN frame corruption	*G102 offset sats CM rejection*	CAN bit errors	U1713-2F + U0100 cascade	*Engine mgmt degraded.*	7	7	6	294	II	*Resolves with CA-01*
G1 strap — terminal faces	High contact resistance	*Residual oxide post-replacement*	Marginal bolt face R	U1960 stored (legacy)	*Minimal current impact.*	8	4	7	224	II	*CA-04 same visit*

Item	Failure Mode	Cause	Local Effect	System Effect	Mission Effect	S	O	D	RPN	Cls	Action
G101 — engine block strap	Partial corrosion	*Al oxide 7yr. Uninspected.*	ECM dual-ground degraded	U1713 may persist post CA-01	*Secondary priority.*	7	4	7	196	III	*CA-05 same visit*
Fuel pump — Recall 93C	Pump failure	*Vitesco defect NHTSA 25V586000.*	Loss of fuel pressure	Engine stall / no-start	*Loss of drive power.*	8	3	6	144	III	*FREE recall at dealer*
PROXI B10AA-4A	Corrupted config flag	*Software artifact of gnd event.*	B10AA-4A in BCM log	No functional impairment	*No mission impact.*	2	9	3	54	IV	*CA-06 AlfaOBD PROXI*
BCM — water intrusion	Connector corrosion	*Pass HVAC scuttle overflow.*	BCM pin corrosion	Multi-module cascade	*ELIMINATED — footwell dry.*	8	1	5	40	III	*Eliminated by evidence*